Test Plan & Quality Evidence

BVA tables, equivalence classes, decision tables, defect log, DRE and defect-density calculations

1. Test Strategy & Visual Quality Summary

ClubSphere's verification strategy combines **7-point Robust Boundary Value Analysis (BVA)** on numeric range fields (`requestedBudgetINR` and `expectedAttendees`), **Equivalence Partitioning & Decision Table Testing** on complex multi-condition domain rules (venue clash detection and 3-tier approval routing), and continuous defect tracking across unit, integration, system, and 46-club pilot UAT phases.

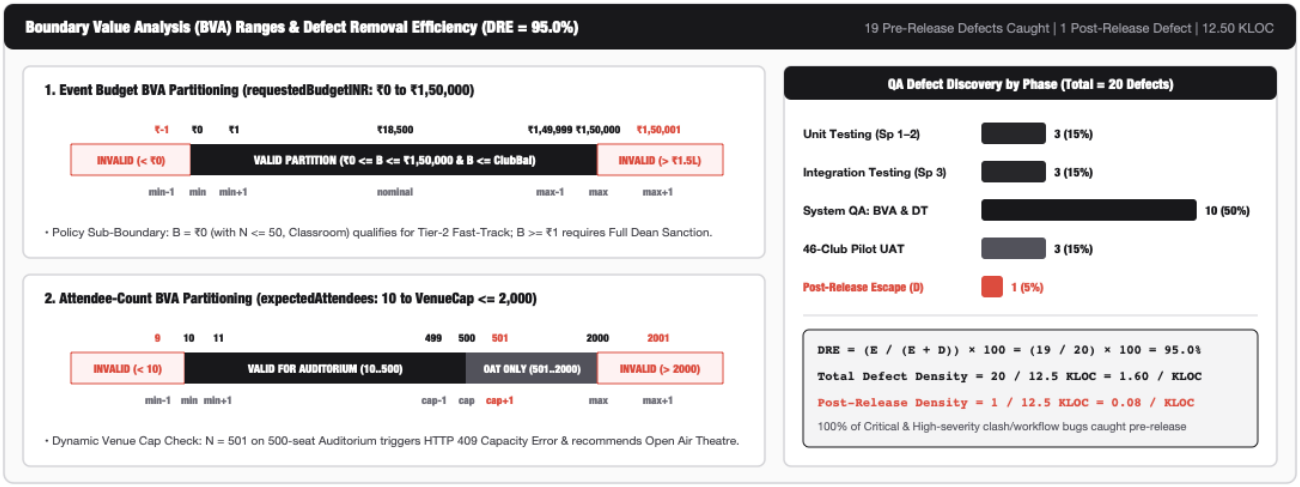


Figure 1 - Visual BVA boundary partitions for Budget and Attendee-Count (left) and QA defect removal funnel achieving 95.0% DRE (right).

2. Boundary Value Analysis (BVA) Tables

2.1 Event Budget Field (requestedBudgetINR: Valid Range Rs. 0 to Rs. 1,50,000)

Test ID	Boundary Point	Input (Rs.)	Precondition (Club Bal)	Expected Outcome	Actual Result
TC-BVA-01	min - 1 (Invalid)	Rs. -1	Rs. 80,000	HTTP 400 Error: Budget cannot be negative (min Rs. 0)	PASS (Rejected)
TC-BVA-02	min (Valid Lower)	Rs. 0	Rs. 80,000	Accepted; flags deanFastTrackEligible=true if N ≤ 50 & Classroom	PASS (Accepted)
TC-BVA-03	min + 1 (Valid)	Rs. 1	Rs. 80,000	Accepted; triggers mandatory Tier-2 Dean sanction & ledger check	PASS (Accepted)
TC-BVA-04	Nominal	Rs. 18,500	Rs. 80,000	Accepted; routes Tier 1 -> Tier 2 -> Tier 3; earmarks Rs. 18,500	PASS (Accepted)
TC-BVA-05	max - 1 (Valid)	Rs. 1,49,999	Rs. 1,50,000	Accepted; within single-event ceiling and club balance	PASS (Accepted)
TC-BVA-06	max (Valid Upper)	Rs. 1,50,000	Rs. 1,50,000	Accepted; exact maximum ceiling allowed for annual flagship event	PASS (Accepted)
TC-BVA-07	max + 1 (Invalid)	Rs. 1,50,001	Rs. 2,00,000	HTTP 422 Error: Exceeds single-event ceiling of Rs. 1,50,000	PASS (Rejected)

2.2 Attendee-Count Field (expectedAttendees: System Range 10..2,000 & Venue Capacity C=500)

Test ID	Boundary Point	Input (N)	Target Venue (Cap)	Expected Outcome	Actual Result
TC-BVA-08	min_sys - 1 (Invalid)	9	Seminar Room (100)	HTTP 400 Error: Minimum attendee count for venue booking is 10	PASS (Rejected)
TC-BVA-09	min_sys (Valid)	10	Seminar Room (100)	Accepted (10 ≤ 100); acquires SOFT_HOLD lock	PASS (Accepted)
TC-BVA-10	min_sys + 1 (Valid)	11	Seminar Room (100)	Accepted (11 ≤ 100); acquires SOFT_HOLD lock	PASS (Accepted)
TC-BVA-11	Cap_venue - 1 (Valid)	499	Main Auditorium (500)	Accepted (499 ≤ 500); acquires SOFT_HOLD lock	PASS (Accepted)
TC-BVA-12	Cap_venue (Exact Cap)	500	Main Auditorium (500)	Accepted (500 ≤ 500 exact fire-safety limit); acquires SOFT_HOLD	PASS (Accepted)
TC-BVA-13	Cap_venue + 1 (Over Cap)	501	Main Auditorium (500)	HTTP 409 Capacity Error; suggests Open Air Theatre (Cap: 2,000)	PASS (Blocked)
TC-BVA-14	max_sys / max_sys + 1	2000 / 2001	Open Air Theatre (2000)	2000 -> Accepted; 2001 -> HTTP 422 Error (Exceeds campus limit 2,000)	PASS (Verified)

3. Equivalence Classes & Decision Tables (Venue Clash & Approval Rules)

3.1 Equivalence Partitions

Input Domain	Valid Equivalence Classes (VEPs)	Invalid Equivalence Classes (IEPs)
Venue Slot & Buffer (FR-03)	EC-V1: Disjoint slot (gap ≥ 30m from any SOFT_HOLD or CONFIRMED slot); EC-V2: Overlaps only RELEASED/CANCELLED slots	EC-V3: Direct overlap with SOFT_HOLD/CONFIRMED; EC-V4: Buffer violation (0–29m gap); EC-V5: Venue in MAINTENANCE
Attendee Count N (FR-02)	EC-N1: Small group (10 ≤ N ≤ 50 & N ≤ Cap); EC-N2: Standard/Large (51 ≤ N ≤ 2,000 & N ≤ Cap)	EC-N3: Under-minimum (N < 10); EC-N4: Exceeds selected venue capacity (N > Cap); EC-N5: Exceeds campus cap (N > 2,000)
Event Budget B & Ledger (FR-05, FR-07, FR-08)	EC-B1: Zero budget (B = Rs. 0); EC-B2: Sanctionable budget (Rs. 1 ≤ B ≤ Rs. 1,50,000 & B ≤ ClubBal & no overdue receipts)	EC-B3: Negative (B < Rs. 0); EC-B4: Exceeds club balance (B > ClubBal); EC-B5: Unsettled past receipts > 7d; EC-B6: B > Rs. 1,50,000

3.2 Combined Decision Table — Venue Clash & Approval-Level Routing Rules

Rule / Test ID	Slot & 30m Buffer Status	Attendees ≤ Cap	Budget & Club Ledger Status	Venue / Risk Profile	→ Expected System Decision & State
R1 (TC-DT-01)	Free (≥ 30m gap)	Yes (N ≤ 50)	B = Rs. 0 (No overdue receipts)	Classroom (Daytime)	Lock SOFT_HOLD -> Tier-1 Faculty -> Fast-Track Tier-2 -> Tier-3 Security -> CONFIRMED
R2 (TC-DT-02)	Free (≥ 30m gap)	Yes (N > 50)	B = Rs. 0 (No overdue receipts)	Auditorium / Evening	Lock SOFT_HOLD -> Full 3-Tier (Faculty -> Dean -> Security) -> CONFIRMED
R3 (TC-DT-03)	Free (≥ 30m gap)	Yes (N ≤ Cap)	Rs. 1 ≤ B ≤ ClubBal (No overdue)	Any valid venue	Lock SOFT_HOLD -> Full 3-Tier + Earmark B on Dean approval -> CONFIRMED
R4 (TC-DT-04)	Overlap SOFT_HOLD / CONFIRMED	Yes	Any valid B	Main Auditorium	REJECT immediately (HTTP 409 Venue Clash) + suggest alternate slots/venues
R5 (TC-DT-05)	Adjacent (< 30m gap: 15m)	Yes	Any valid B	Any venue	REJECT immediately (HTTP 409 Buffer Violation) + suggest +15m shifted start
R6 (TC-DT-06)	Free (≥ 30m gap)	No (N > Cap)	Any valid B	Seminar Hall (N=350 > 200)	REJECT immediately (HTTP 409 Capacity Exceeded) + suggest Auditorium
R7 (TC-DT-07)	Free (≥ 30m gap)	Yes	B > ClubBal OR Overdue > 7d	Any venue	BLOCK submission (HTTP 422 Insufficient Funds / Unsettled Past Event Receipts)
R8 (TC-DT-08)	Free (≥ 30m gap)	Yes	Rs. 1 ≤ B ≤ ClubBal	Rejected by Tier 1/2/3	Transition to REJECTED -> emit ProposalRejectedEvent -> unfreeze B & RELEASE slot

4. Defect Log & Quality Metrics (DRE & Defect Density)

Defect ID	Phase Found	Module	Severity	Defect Description & Root Cause	Fix & Verification Test
DEF-01..03	Unit (Sp 1–2)	Proposal / Venue	High/Med	DEF-01: Negative float -0.01 accepted; DEF-02: Whitespace title; DEF-03: `<` instead of `<=` rejected exact capacity N=500	Switched to Decimal >= 0, trim check, and `attendees <= capacity` (TC-BVA-01, TC-BVA-12)
DEF-04..06	Integration (Sp 3)	VenueSlotBooking	Critical/High	DEF-04: 5ms race condition allowed dual SOFT_HOLD; DEF-05: 30m buffer checked only post-event; DEF-06: Suggested MAINTENANCE room	Added PostgreSQL GiST exclusion index, symmetric ±30m buffer, and ACTIVE filter (TC-03, TC-DT-04..05)
DEF-07..10	System QA (Sp 4–5)	ApprovalWorkflow	Critical/High	DEF-07: Revision released SOFT_HOLD early; DEF-08: B=Rs. 0 Auditorium skipped Dean; DEF-09: Security reject orphaned SOFT_HOLD; DEF-10: Mobile overflow	Added 24h revision hold, compound fast-track guard, event bus listener bind, flex-wrap (TC-04..06)
DEF-11..16	System QA (Sp 6–8)	Ledger / Cert / Audit	High/Med	DEF-11: Concurrent earmark overdraft; DEF-12: MIME bypass; DEF-13: Holiday count in 7d gate; DEF-14: CSV BOM crash; DEF-15: Handover JWT; DEF-16: Audit DELETE	Added FOR UPDATE lock, magic-byte check, holiday table, utf-8-sig, JWT blacklist, REVOKE DELETE (TC-07..14)
DEF-17..19	UAT Pilot (Wk 17–18)	SLA / UI / Export	Med/Low	DEF-17: Duplicate 16h SLA emails; DEF-18: SOFT_HOLD badge contrast 3.8:1; DEF-19: 5-year export took 2.4s (>2.0s NFR-04)	Added idempotency flag, darkened badge text to #713f12, added composite DB index (0.64s) (TC-11, 15, 17)
DEF-20	Post-Release (Wk 23)	CalendarExport	Low	DEF-20 (Escaped): .ics calendar invite lacked explicit TZID=Asia/Kolkata in desktop Outlook	Patched VTIMEZONE block in .ics generator within 4 hours

4.1 Worked Calculations for Defect Removal Efficiency (DRE) & Defect Density

- **Delivered Codebase Size: 6.80 KLOC (Backend/SQL) + 5.70 KLOC (React Frontend) = 12.50 KLOC** across **96 Story Points**.
- **Pre-Release Defects Caught (E): E = 19 defects** (3 Unit + 3 Integration + 10 System QA + 3 UAT Pilot).
- **Post-Release Latent Defects (D): D = 1 defect** (DEF-20 minor .ics timezone issue).
- **Defect Removal Efficiency (DRE): DRE = (E / (E + D)) × 100% = (19 / (19 + 1)) × 100% = (19 / 20) × 100% = 95.00%**.
- **Total Lifecycle Defect Density: DD_total = (E + D) / Size = 20 / 12.50 KLOC = 1.60 defects / KLOC** (or **20 / 96 = 0.208 defects / Story Point**).
- **Post-Release (Escaped) Defect Density: DD_post = D / Size = 1 / 12.50 KLOC = 0.08 defects / KLOC** (0 Critical/High escapes).